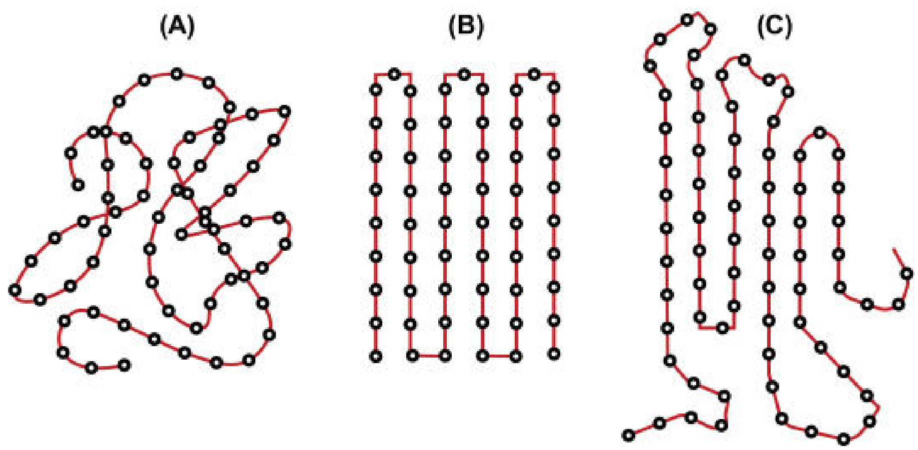
October 26, 2022

APS110 – Lecture 20

See hand written notes for drawings.

Crystallinity in polymers

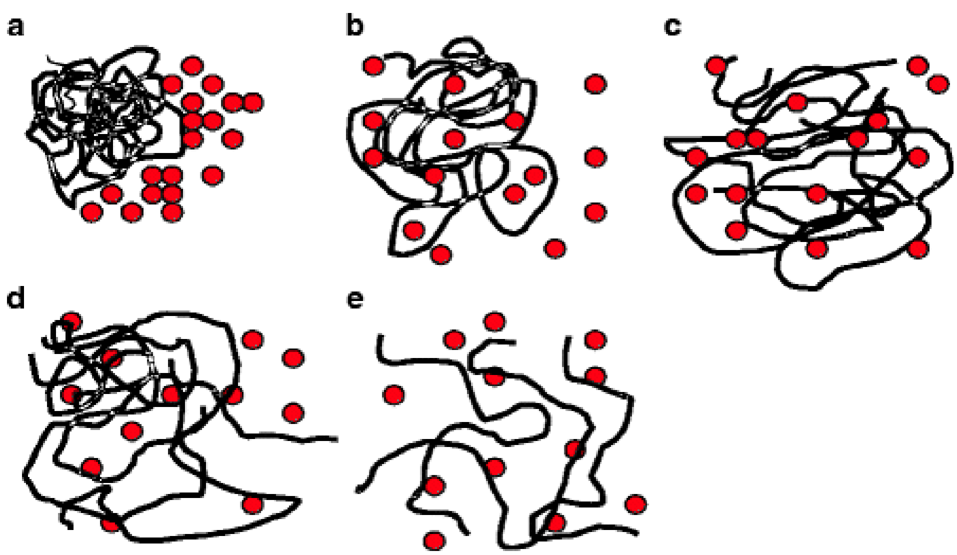
- Different from crystallinity in metals and ceramics → these materials can be completely crystalline
  - Polymers can never be completely crystalline
  - Why? Just statistics → the molecules are soooo long that it's hard for the entire sample to have polymer molecules folded in such an organized way
  - Polymers will have areas of crystallinity that are surrounded by amorphous regions (we can call these 'tie molecules')
  - We can do things to encourage crystallinity, but we'll only be able to achieve ~80% crystallinity
- We can have a completely amorphous polymer *[see (A)]*
  - Ex. acrylic is an amorphous polymer



- Which will be stronger: entirely amorphous or highly crystalline?
  - Which will have a higher yield strength?
  - In polymers, plastic deformation involves sliding molecules past each other
  - So, what is holding the molecules together – what is resisting the deformation?
  - In a crystalline sample, there are a LOT of inter-molecular interactions (secondary bonding) between the molecules that are holding the molecules together
  - In an amorphous sample, there are much fewer of these secondary bonds
- **Semi-crystalline samples will have a <u>higher</u> yield strength, and will have a <u>higher</u> elastic constant**
  - **Elastic constant** → *essentially* Young's modulus
  - Unlike how prof has stressed previously, that Young's modulus is structure INDEPENDENT, the elastic constant changes with changes in microstructure
  - Young's modulus should relate to a **single bond type**

- The **elastic constant is a <u>composite</u> behaviour** between C-C bonds, C-H bonds, C-other bonds, secondary bonds and interactions
  - Composite of bond behaviour from several bond types
- You *can* use Young's modulus to describe polymers, but technically it is not the most accurate description of polymer behaviour

- Strength, resistance to chemical dissolution, melting/heating, are all very similar mechanisms at molecular level
  - **Dissolving** – solvent molecule (small) needs to migrate into the structure and break the secondary bonds holding the sample together
    - We want to separate one molecule from its neighbours by surrounding it with solvent molecules
    - So which will be harder to dissolve – amorphous or semi-crystalline?
      - Semi-crystalline, since there are more secondary bonds and interactions holding the molecules together
      - Also, the molecules are more tightly packed and so it's physically harder for the solvent molecules to migrate into the crystalline region



  - **Heating**
    - At higher temperatures, the molecules have more kinetic energy and start vibrating more
    - At what point will the molecules have enough kinetic energy to overcome the secondary interactions (weak forces/bonds)
    - Amorphous → there is more space within structure, molecules aren't as close together (lower density)
      - Smaller amounts of kinetic energy will allow the molecules to move away from one another

    - Semi-crystalline → the molecules are packed very closely, and so it takes a much higher temperature for the molecules to have sufficient energy to move away from one another

Chain Length

- How do we measure length?
  - How do we quantify this, what are the units?
  - The molecule isn't straight, it's not easy to 'measure' in the classic way
  - Number of repeating units? → sometimes we use this
  - **Molecular weight (g/mol)**